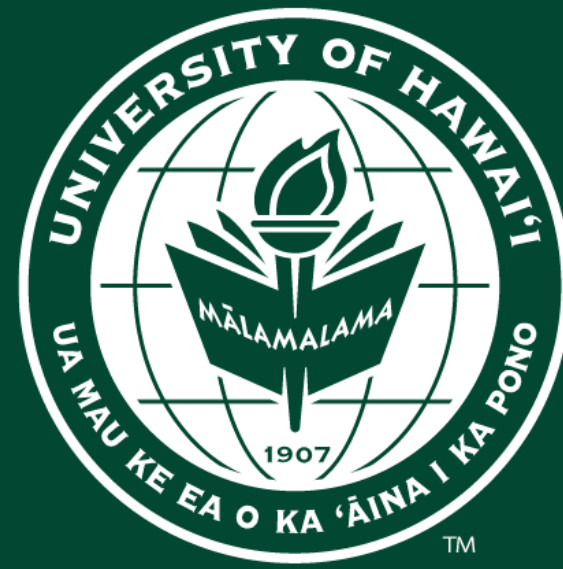




ShepBERT: An OMOP-Portable EHR Transformer Pipeline for Multi-Cancer Risk Prediction in NHPI-Serving Health Systems

Pavel Bushuyeu^{1,2} • Igor Molybog, PhD¹ • John Shepherd, PhD^{1,2}

¹University of Hawai'i at Mānoa • ²University of Hawai'i Cancer Center



UNIVERSITY
of HAWAI'I
MĀNOA

Background & Motivation

NHPI Subpopulation — Hidden Disparities

NHPI cancer burden is high, but NHPI patients remain underrepresented in EHR AI. ShepBERT is an OMOP-portable transformer with built-in leakage control, ready for local validation.

175 vs 124

Breast incidence /100K (NHPI vs US)

22% vs ~9%

Stage III+ breast cancer Dx (Hawai'i vs contiguous US)

4

Target cancers (breast, colorectal, lung, prostate)

Stage III+ breast cancer: **22%** of Hawai'i diagnoses vs **~9%** mainland — and **50–80%** in some Pacific settings with limited screening.

Aggregate HI Rates Look Unremarkable — Masking NHPI Burden

Cancer	HI Inc.	US Inc.	HI Mort.	US Mort.
Breast	140.2	127.0	15.9	19.6
Prostate	101.1	110.5	14.9	18.8
Lung	41.7	54.0	26.6	35.0
Colon & Rectum	38.8	36.5	11.8	13.1

Statewide rates sit at or below US averages — the NHPI disparity hides in subgroups (above), invisible in aggregates.

Per 100k; HI Tumor Registry & SEER, 2012–2016.

Clinical AI Gap

- Aggregate rates mask NHPI burden, stage, and screening gaps.
- Mainland-trained clinical AI may not be locally valid for NHPI patients.

Our Approach

- Pre-train on MIMIC-IV masked EHR codes.
- Fine-tune 1-year risk for four cancers.
- Transfer through OMOP to Queen's Epic for validation.

Prediction Task

Prediction row	one per patient-year
Input window	EHR history before prediction age
Target	first cancer within 1 year
Negatives	≥18 mo cancer-free follow-up
Split	patient-level 5-fold CV

Methods

Data Pipeline



Vocabulary Reduction via OMOP Harmonization

36,628 raw concept codes	26,431 OMOP concepts	-28% vocabulary reduction
------------------------------------	--------------------------------	-------------------------------------

Feature Encoding

Input	Example	Captures
Diagnosis / procedure	OMOP concept	clinical history
Age	47.3 years	age-related risk
Time offset	+821 days	disease progression
Visit segment	visit 3	care episode
Demographics	race / sex tokens	population context

Pre-training Objective

Hide 15% of EHR tokens and predict them from context, learning reusable sequence structure without cancer labels.

How ShepBERT Works



Model Architecture

Transformer layers	6
Attention heads	12
Hidden dimension	288
Max sequence length	512 tokens
Total parameters	~5 M
Base architecture	CORE-BEHRT

Three-Pass Leakage Audit

Censors cancer-pathway shortcut codes:

- Pass 1:** seed list for Dx, chemo, staging, and radiation.
- Pass 2:** >100× case/control enrichment screen.
- Pass 3:** EG-SHAP audit for screening proxies.

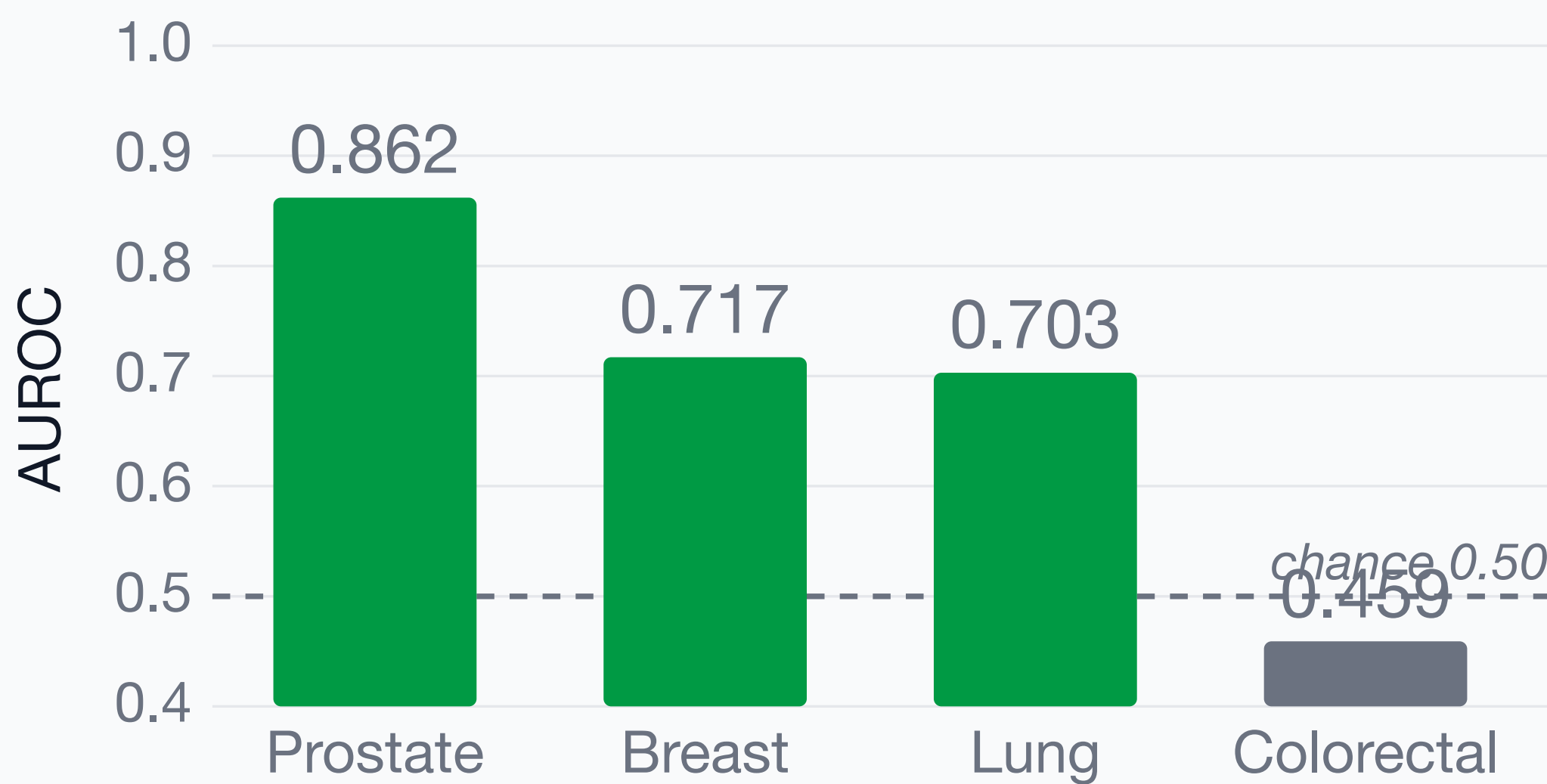
What We Validated So Far

Pre-training — MIMIC-IV analytic cohort (230,948 patients)

3.630 Best validation loss (epoch 20)	9.5 hrs Training time RTX 5000 16 GB
22.2% Exact hidden-code prediction accuracy	60.5% Correct within 10 guesses
Chance is <0.01% over 26,000+ codes. 22.2% exact $\approx$ 1-in-5; 60.5% top-10 $\approx$ 3-in-5 — on par with BEHRT benchmarks.	

Synthetic Validation: Signal and Leakage Checks

20K Synthea patients (Hawai'i demographics) • 5-fold patient-level CV • four first-cancer tasks.



Attribution as a Leakage Check

IG + EG-SHAP flagged colorectal screening proxies pre-censoring. After censoring, colorectal AUROC → **0.459**: shortcut removed, not real signal.

Rare-Event Handling

Prevalence only **~0.03–0.06%** → few positives per batch. Square-root inverse-frequency sampling oversamples without collapsing classes.

How Fine-Tuning Works

Pre-trained encoder → mean-pool the patient timeline → predict first cancer within 12 months.

What This Validates

Pipeline runs end-to-end with leakage censoring. Synthea isn't NHPI-stratified — Queen's Epic is the clinical validation cohort. Colorectal → chance after censoring: the cleanest leakage stress test.

Next Step: Queen's Epic Validation

Queen's Medical Center (Honolulu) provides longitudinal Epic EHR depth for NHPI-specific validation across local diagnoses, medications, labs, procedures, and imaging.

Queen's Epic → **OMOP** → **ShepBERT** → **Fine-tune + calibrate**

- Validate on de-identified Queen's Epic data (DUA in progress).
- Test **local validity** through the same OMOP workflow.
- Report NHPI-stratified discrimination, calibration, and attribution.
- Extend context length with a causal sequence model.

Conclusion

- What it is:** an OMOP-standardized EHR transformer for 1-year multi-cancer risk with a three-pass leakage audit.
- Validated end-to-end:** pre-training, fine-tuning, and shortcut censoring across four cancers on public and synthetic data.
- Next:** NHPI-specific validation and calibration on Queen's Epic.



Questions, Ideas & Updates

Scan to ask questions, share ideas, or request release updates.

References

- Odgaard et al. CORE-BEHRT. arXiv, 2024.
- Li et al. BEHRT. Sci. Reports, 2020.
- Devlin et al. BERT. NAAACL, 2019.
- Rasmy et al. Med-BERT. npj Digit. Med., 2021.
- Steinberg et al. CLMBR. J. Biomed. Inform., 2021.
- Johnson et al. MIMIC-IV. Sci. Data, 2023.
- Walonoski et al. Synthea. JAMIA, 2018.
- OHDSI. OMOP Common Data Model, 2019.
- Sundararajan et al. Integrated Gradients. ICML, 2017.
- Erion et al. Expected Gradients. Nat. Mach. Intell., 2021.
- Deng et al. Breast cancer burden in Hawai'i. CEBP, 2018.
- Sarfati et al. Cancer control in the Pacific. Lancet Oncol., 2019.